Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/06/2009
PI Name:	Shaw, Benjamin Ph.D.		
Project Title:	Reduced Gravity Studies of Soret Transport Effects in Liquid Fuel Combustion		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2003	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2008 per PI (10/07) NOTE: end date changed to 9/30/2007 per PI and GRC (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	Soret transport, which is mass transport driven by thermal gradients, can be important in practical flames as well as laboratory flames by influencing transport of low molecular weight species (e.g, He, H and H2) and high molecular weight fuel species (e.g., C7H16) and O2. This research is concerned with investigating effects of Soret transport on combustion of liquid fuel droplets. Experiments will involve normal-gravity and reduced-gravity experiments on combustion of liquid fuel droplets in environments where Soret effects significantly influence transport of fuel and oxygen to flame zones. The experiments will employ pressures ranging from 0.3 to 10 atm, with extensions to 30 atm also being considered. Experiments will also be performed where Soret effects are not expected to be important. Droplets initially in the 0.5 to 1 mm size range will be burned. Data will be obtained on influences of Soret transport on combustion characteristics (e.g., droplet burning rates, droplet lifetimes, flame extinction, and transient flame behaviors) under simplified geometrical conditions that are most amenable to theoretical modeling (i.e., spherical symmetry). The experimental results will be compared with existing theoretical models as well as new models that will be developed as part of this program. The knowledge gained will be of use in evaluating combustion mechanisms under reduced gravity, which is of interest for spacecraft fire safety. It is noted that in fall 2005, the PI became involved with a different program that includes a team of investigators from various universities as well as from NASA. This program, which is termed the Flammability and Extinction Experiment (FLEX), is focused upon using droplets to study flammability and extinction under reduced-gravity conditions of relevance to spaceflight and human exploration of space. The FLEX program is both experimental and theoretical in nature. Experiments presently include reduced-gravity studies as well as normal-gravity comparison experiments. In the research, individual droplets are dispensed, deployed and ignited in reduced gravity (as well as normal gravity). Theoretical approaches include developing computational and analytical models, with a particular emphasis on evaluating influences of gas-phase diffusive transport as well as droplet internal flows, which may arise from Marangoni stresses as well as droplet deployment and ignition. The present research on influences of Soret transport on droplet combustion is strongly complementary to certain aspects of the FLEX program, e.g., research on influences of gas-phase transport properties on flammability and extinction of droplets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has the potential to produce new and useful information on influences of Soret transport combustion mechanisms, both in normal gravity as well as reduced gravity. For example, Soret transport can be important in low-gravity flames by significantly influencing transport of O2 to a flame zone when low molecular weight (e.g., He) or high molecular weight (e.g., SF6) species are present in the gas phase. Low gravity combustion characteristics such as flame sizes, flame temperatures and sooting can be significantly influenced by Soret transport. Increasing knowledge about these effects is important for applications such as spacecraft fire safety. Soret transport can also influence transport of high molecular weight fuel species, which are commonly employed in terrestrial engines. The present research has the potential to provide new information that will increase understanding of combustion mechanisms that are relevant to terrestrial applications that involve use of liquid fuels.
Task Progress:	Work has principally involved performing normal-gravity and reduced-gravity experiments, analyzing data from these experiments and performing theoretical analyses. Summaries of the progress are given below. Normal-Gravity and Reduced Gravity Experiments: Reduced gravity and normal gravity combustion experiments were performed with individual fiber-supported droplets that were initially about 1 mm in diameter. These experiments were performed with a goal of investigating influences of Soret transport and multicomponent diffusion on droplet flammability characteristics. Droplets composed of heptane, methanol or propanol were investigated and a thermocouple and radiometer were employed in conjunction with video cameras to obtain data on flammability behavior. The environment was composed of an air-inert mixture at about 298 K, where helium and carbon dioxide were separately used as inerts. The pressure was varied from 0.3 to 3 MPa. In the experiments, the amount of inert gas in the environment was progressively increased until droplets could no longer be ignited. Droplet ignitability depended on the gravitational level as well as the inert gas in the environment. Analysis of the data is still in progress, but in general, the amount of inert gas required to suppress combustion was higher in reduced gravity than normal gravity. Other: The gas temperature is predicted to be strongly influenced by Soret effects under certain circumstances, e.g., when xenon or helium are used as a diluent, and as a result it is useful to be able to measure the gas temperature in a non-intrusive manner. Efforts were directed towards development of a simple low-power diode laser system than could be potentially used on the UCD drop rig for measurement of gas temperature and concentrations of selected species, e.g., CO or H2O, via tunable diode laser absorption spectroscopy (TDLAS). A small lock-in amplifier composed of only a few electrical components was constructed and tested, and the amplifier was able to detect the 2f output from a modulated diode laser under laboratory conditions. Software-based signal processing options were also investigated in terms of analysis of TDLAS signals. Specifically, harmonic wavelets were investigated and found to be useful for extracting harmonics from a modulated diode laser signal.
Bibliography Type:	Description: (Last Updated: 04/29/2010)
Articles in Peer-reviewed Journals	Shaw BD, Wei JB. "Propanol droplet flammability and combustion in air-diluent environments under normal and reduced gravity." Combustion Science and Technology, 2007 Jun;179(6):1205-23. http://dx.doi.org/10.1080/00102200601057394 , Jun-2007
Articles in Peer-reviewed Journals	Duan H, Gautam A, Shaw BD, Cheng HH. "Harmonic wavelet analysis of modulated tunable diode laser absorption spectroscopy signals." Applied Optics 2009 Jan 10;48(2):401-7. http://dx.doi.org/10.1364/AO.48.000401 , Jan-2009
Articles in Peer-reviewed Journals	Wei JB, Shaw BD. "Reduced gravity combustion of propanol droplets in oxygen-inert environments." Combustion Science and Technology, 2009 Dec;181(12):1480-94. http://dx.doi.org/10.1080/00102200903190851 , Dec-2009

Papers from Meeting Proceedings	Shaw BD, Wei, JB. "Influence of Gravity on Combustion and Flammability of Heptane Droplets in Air-Diluent Environments." Paper G17 presented at the Fifth US Combustion Meeting, University of California at San Diego, La Jolla, CA, March 25-28, 2007. Paper G17, Fifth US Combustion Meeting, University of California at San Diego, La Jolla, CA, March 25-28, 2007. , Mar-2007